47202 I.F.E. Problems, Nuclear 2026
Solutions

Calculations

1. Fission of one U-235 atom releases about 200 MeV of energy. How many kWh thermal energy is released from fission of 1g ^{235}U?

q_{atom} = 200 MeV atom^{-1}

M = 235 g mol^{-1}

m = 1 g

N_A = 6.023 · 10^{23} mol^{-1}

$$Q = q_{atom} \cdot n_{atoms} = q_{atom} \cdot n_{mole\ atoms} \cdot N_A = q_{atom} \cdot \frac{m}{M} \cdot N_A$$

$$= 200\ MeV\ atom^{-1} \frac{1\ g}{235\ g\ mol^{-1}} 6.022 \cdot 10^{23}\ \text{mol}^{-1} = \underline{5.125 \cdot 10^{23}\ \text{MeV}}$$

Converted to joule:

Q = 5.125 · 10^{23} · 10^{6} · 1.602 · 10^{-19} J eV^{-1} = 8.210 · 10^{10} J = <u>82.1 GJ</u>

Or <u>22.8 MWh</u>

2. The average uranium concentration in Earth's crust is 2.8 ppm. In typical mines the concentration ranges from 0.02% to 0.4% (determined as U_3O_8), although at certain locations it can be up to 20%. If mined from a deposit with 0.4% uranium (U_3O_8). How much ore must ideally be process to charge an electric vehicle battery of 75 kWh (assume that only U-235 is utilized, electricity is produced with a carnot efficiency of 40%, and that there are no losses in the entire process)

75 kWh = 75 · 10^3 Wh · 3600 s h^{-1} = 2.70 · 10^8 J = <u>270 MJ</u>

Primary energy needed: 270 MJ / 0.40 = <u>675 MJ</u>

A gram of U-235 gives 82.1 GJ, so 675 MJ/82100 MJ g^{-1} = <u>0.0082 g</u>

In U_3O_8 the uranium content is $\frac{3 \cdot 238\ g\ mol^{-1}}{3 \cdot 238\ g\ mol^{-1} + 8 \cdot 16\ g\ mol^{-1}}$ = <u>85%</u>

Uranium concertation in ore = <u>0.4%</u>

U-235 the content in natural uranium = 0.7%

Concentration of U-235 = 85% · 0.4% · 0.7% = 0.0024%

The amount of ore to process is thus 0.0082 g / 0.000024 = <u>341 g ore</u>

Solutions

3. According to IAEA, a liter of sea water contains about 33 mg of deuterium. How much fusion energy does that correspond to? (the energy of fusion is 17.6 MeV).

0.033 g deuterium equals 0.0165 moles.

$n_D = \frac{m}{M} N_A = \frac{0.033\ g}{2\ g\ mol^{-1}} 6.022 \cdot 10^{23}\ mol^{-1} =$ 9.94·10^{21} atoms

Energy released = 9.94·10^{21} atoms · 17.6 MeV = 1.75 ·10^{23} MeV

= 1.75 ·10^{29} eV

Converted to joule: Energy released = 1.75 ·10^{29} eV · 1.602 · 10^{-19} J eV^{-1}

2.80 · 10^{10} J = 28 GJ

Or ca. 7.8 MWh

For comparison, one liter of crude oil contains 35-40 MJ or ca. 10-12 kWh

4. In a BWR or PWR, steam is generated with a temperature of about 290 °C. (a) If river water used to receive excess heat has a temperature of 20 °C, what is the maximum possible conversion efficiency of the reactor's thermal energy into electricity? (b) Nuclear power plants typically have conversion efficiencies of 34 %. Why is this efficiency less than the ideal efficiency?

(a) $\eta = \frac{T_h - T_l}{T_h} = \frac{290-20}{563.15} = 0.48 =$ 48%

(b) Real sytems are always less efficeint than what thermodynamics predict som the absolut maximum. One way to increase efficiency is to genrate heat at a higher temperature.

5. A 1000 MWe nuclear power plant has a thermal conversion efficiency of 33%. (a) How much thermal power is rejected through the condenser to the cooling water? (b) What is the flow rate (kg/s) of the condenser cooling water if the temperature rise of this water is 12 °C? Specific heat of water is 4184 J kg^{-1}K^{-1}.

(a) $W_e = Q_{total} \cdot 0.33 \iff = Q_{total} = \frac{W_e}{0.33}$

$Q_{heat} = Q_{total} - W_e = \frac{W_e}{0.33} - W_e = W_e \frac{1-0.33}{0.33} = 1000\ \text{MW} \frac{1-0.33}{0.33} =$ 2030 MW

(b) $\dot{m} = \frac{Q_{heat}}{C_v \Delta T} = \frac{2030 \cdot 10^6\ \text{J s}^{-1}}{4184\ \text{J kg}^{-1}\text{K}^{-1} \cdot 12\ \text{K}} =$ 40432 kg s^{-1}

This is 40 m^3 s^{-1}

6. The energy of U-235 fusion is about 200 MeV. What is the energy of combustion of a methane molecule in the unit eV? A heat of combustion table can be found on Learn.

Heat of combustion: Heat of combustion: 890.8 kJ mol^{-1}

The energy of one molecule is $\frac{890.8\ \text{kJ mol}}{N_A} = \frac{890{,}800\ \text{J mol}^{-1}}{6.022 \cdot 10^{23}\ \text{mol}^{-1}} =$ $1.479 \cdot 10^{-18}$ J

Conversion to eV: $1.479 \cdot 10^{-18}$ J / $1.602 \cdot 10^{-19}$ J eV^{-1} = 9.23 eV

This is about 20,000,000 times less than the fission energy of U-235

7. Calculate the mass defect of $^{232}_{90}\text{Th}$. The atomic mass is 232.038 u. Find additional information needed in the list of formulas.

90 protons or 90 protium:

(232-90) neutrons = 142 neutrons:

Mass of constituent: 90·1.007825 u + 142·1.008665 = 233.935 u

Mass defect: 233.935 u - 232.038 u = 1.897 u